

Sample Problems and Solutions
Discrete Mathematics
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[Propositional Logic.]

1. [10 points] Give negations of the following statements using formal mathematical language. Your negations must not use conditionals or biconditionals.
- (a) [1 point] if p , then q
 - (b) [1 point] p only if q
 - (c) [1 point] p if and only if q
 - (d) [1 point] p is necessary for q
 - (e) [1 point] p is sufficient for q
 - (f) [1 point] p is necessary but not sufficient for q
 - (g) [1 point] p is not necessary but sufficient for q
 - (h) [1 point] p is necessary and sufficient for q
 - (i) [2 points] p is neither necessary nor sufficient for q

Solution.

- (a) $p \wedge \sim q$
 - (b) $p \wedge \sim q$
 - (c) $p \oplus q$
(alternate answer) $(p \wedge \sim q) \vee (\sim p \wedge q)$
 - (d) $q \wedge \sim p$
 - (e) $p \wedge \sim q$
 - (f) $(q \wedge \sim p) \vee (\sim p \vee q)$
 - (g) $(\sim q \vee p) \vee (p \wedge \sim q)$
 - (h) $p \oplus q$
(alternate answer) $(q \wedge \sim p) \vee (p \wedge \sim q)$
 - (i) $(\sim q \vee p) \vee (\sim p \vee q) \equiv \mathbf{t}$
2. [5 points] Determine if the following deduction rule is valid:
- $$\begin{array}{l} p \rightarrow q \\ \sim q \rightarrow \sim r \\ \hline \end{array}$$

$$\therefore p \rightarrow r$$

Solution.

p	q	r	$p \rightarrow q$	$\sim q$	$\sim r$	$p \rightarrow q$	$\sim q \rightarrow \sim r$	$p \rightarrow r$
T	T	T	T	F	F	T	T	T
T	T	F	T	F	T	T	T	F
T	F	T	F	T	F	F	F	
T	F	F	F	T	T	F	T	
F	T	T	T	F	F	T	T	T
F	T	F	T	F	T	T	T	T
F	F	T	T	T	F	T	F	
F	F	F	T	T	T	T	T	T

The deduction rule is invalid.

3. [5 points] Determine if the following deduction rule is valid:

$$\begin{array}{l} p \vee q \\ \sim p \vee \sim q \\ \hline \end{array}$$

$$\therefore p \oplus q$$

Solution.

p	q	$p \vee q$	$\sim p$	$\sim q$	$\sim p \vee \sim q$	$p \oplus q$
T	T	T	F	F	F	
T	F	T	F	T	T	T
F	T	T	T	F	T	T
F	F	F	T	T	T	

The deduction rule is valid.

4. [5 points] Determine if the following deduction rule is valid:

$$\begin{array}{l} p \rightarrow \sim q \\ p \wedge q \\ \hline \end{array}$$

$$\therefore p \vee q$$

Solution.

p	q	$\sim q$	$p \rightarrow \sim q$	$p \wedge q$	$p \vee q$
T	T	F	F	T	
T	F	T	T	F	
F	T	F	T	F	
F	F	T	T	F	

The deduction rule is valid (vacuous truth).

5. [5 points] Determine the validity of the following statements.

- (a) [1 point] $p \wedge \sim p \equiv \mathbf{t}$
- (b) [1 point] $p \rightarrow p \equiv \mathbf{t}$
- (c) [1 point] $p \oplus q \equiv (\sim p \vee q) \wedge (\sim q \vee p)$
- (d) [1 point] $p \rightarrow (q \rightarrow r) \equiv (p \rightarrow q) \rightarrow r$
- (e) [1 point] $(p \wedge q \wedge \sim r) \vee p \equiv p$

Solution.

- (a) False ($p \wedge \sim p \equiv \mathbf{c}$)
- (b) True
- (c) False ($p \oplus q \equiv (\sim p \wedge q) \vee (\sim q \wedge p)$)
- (d) False (Implication is not associative)
- (e) True

6. [5 points] Determine how many ordered truth assignments satisfy the following statement:

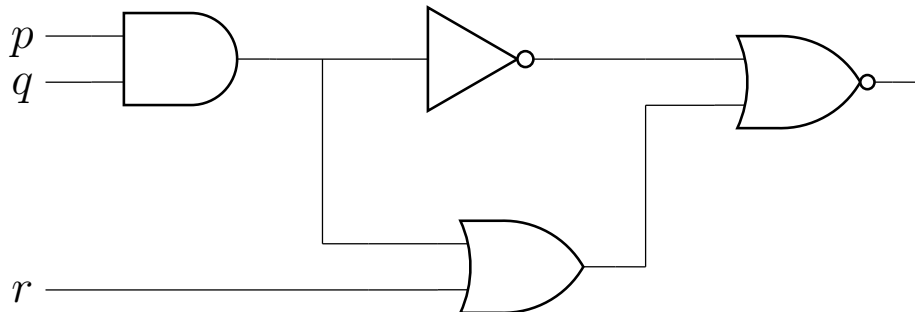
$$\sim (p \wedge q \wedge r) \vee (p \wedge \sim q \wedge r) \vee \sim p$$

Solution.

p	q	r	$p \wedge q \wedge r$	$\sim (p \wedge q \wedge r)$	$p \wedge \sim q \wedge r$	$\sim p$	Final Expression
T	T	T	T	F	F	F	F
T	T	F	F	T	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	F	T	F	T	T
F	T	F	F	T	F	T	T
F	F	T	F	T	F	T	T
F	F	F	F	T	F	T	T

7 truth assignments.

7. [5 points] Given the following logic circuit, construct an Input-Output table.



Solution.

p	q	r	$p \wedge q$	$\sim (p \wedge q)$	$(p \wedge q) \vee r$	$\sim (\sim (p \wedge q) \vee ((p \wedge q) \vee r))$
1	1	1	1	0	1	0
1	1	0	1	0	1	0
1	0	1	0	1	1	0
1	0	0	0	1	0	0
0	1	1	0	1	1	0
0	1	0	0	1	0	0
0	0	1	0	1	1	0
0	0	0	0	1	0	0

8. [5 points] Construct the truth table for the following logical expression:

$$((p \rightarrow q) \wedge r) \leftrightarrow \sim (p \oplus q)$$

Solution.

p	q	r	$p \rightarrow q$	$(p \rightarrow q) \wedge r$	$p \oplus q$	$\sim (p \oplus q)$	$(p \rightarrow q) \wedge r \leftrightarrow \sim (p \oplus q)$
T	T	T	T	T	F	T	T
T	T	F	T	F	F	T	F
T	F	T	F	F	T	F	T
T	F	F	F	F	T	F	T
F	T	T	T	T	T	F	F
F	T	F	T	F	T	F	T
F	F	T	T	T	F	T	T
F	F	F	T	F	F	T	F

9. [5 points] Let us define a new operator $\diamond$ with the following truth table:

p	q	$p \diamond q$
T	T	T
T	F	T
F	T	F
F	F	T

Let us extend another of de Morgan's laws as shown below. Determine whether this property holds for the operator $\diamond$:

$$p \diamond (q \wedge r) \stackrel{?}{=} (p \diamond q) \vee (p \diamond r)$$

Solution.

p	q	r	$q \wedge r$	$p \diamond q$	$p \diamond r$	$p \diamond (q \wedge r)$	$(p \diamond q) \vee (p \diamond r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	F	F	F	F
F	T	F	F	F	T	T	T
F	F	T	F	T	F	T	T
F	F	F	F	T	T	T	T

Yes, the property holds.

10. [2 points] Analyze the following logical statement:

If I study for this exam and attend class, then I will pass the class.

Which of the following statements are logically equivalent to the given statement?

- (a) If I pass the class, then I studied for this exam and attended class.
- (b) If I pass the class, then I studied for this exam or attended class.
- (c) If I do not pass the class, then I did not study for this exam and did not attend class.
- (d) If I do not pass the class, then I did not study for this exam or did not attend class.
- (e) I did not pass the class because I did not study for this exam and did not attend class.

Solution. (d)

11. [2 points] Consider the following logical expression:

$$\sim (P \rightarrow Q) \wedge (R \rightarrow (P \vee Q)).$$

How many possible truth assignments make this expression evaluate to true?

- (a) 0
- (b) 1
- (c) 2
- (d) 3
- (e) 4

Solution. (c)

12. [2 points] Identify the logical reasoning used in the reasoning below:

I know that if it rains, the ground will be wet, and since the ground is wet, it must have rained.

- (a) Modus Ponens
- (b) Modus Tollens
- (c) Converse Error
- (d) Inverse Error
- (e) Generalization

Solution. (c)

13. [2 points] Identify the logical reasoning used in the reasoning below:

Since it is true that Elizabeth and I passed, it is also true that I passed.

- (a) Specialization
- (b) Generalization

- (c) Modus Ponens
- (d) Modus Tollens
- (e) Division into Cases

Solution. (a)

14. [2 points] Identify the logical reasoning used in the reasoning below:

Since I study logic, it is also true that I study logic or art.

- (a) Specialization
- (b) Generalization
- (c) Modus Ponens
- (d) Modus Tollens
- (e) Division into Cases

Solution. (b)

15. [2 points] Identify the logical reasoning used in the reasoning below:

If I study art, I will pass this class, but if I do not study art, I will still pass this class, therefore, I will pass this class regardless.

- (a) Specialization
- (b) Generalization
- (c) Modus Ponens
- (d) Modus Tollens
- (e) Division into Cases

Solution. (e)

[Predicate Logic.]

1. [5 points] Negate the following statements using formal predicate logic:

- (a) [1 point] $\forall x \in \mathbb{R}, P(x) \implies Q(x)$
- (b) [1 point] $\exists x \in \mathbb{R}, P(x) \wedge Q(x)$
- (c) [1 point] $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x^2 + y^2 = 1$
- (d) [1 point] $\forall x, y \in \mathbb{Z}, x \neq y \implies P(x) \wedge P(y)$
- (e) [1 point] $\exists x \in \mathbb{N}, P(x) \vee Q(x)$

Solution.

- (a) $\exists x \in \mathbb{R}, P(x) \wedge \sim Q(x)$
- (b) $\forall x \in \mathbb{R}, \sim P(x) \vee \sim Q(x)$
- (c) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x^2 + y^2 \neq 1$
- (d) $\exists x, y \in \mathbb{Z}, (x \neq y) \wedge (\sim P(x) \vee \sim P(y))$
- (e) $\forall x \in \mathbb{N}, \sim P(x) \wedge \sim Q(x)$

2. [5 points] Consider a scenario where a student is represented by S , and the following predicates are given:

- $C(S)$: Returns true if the student studies computer science.
- $M(S)$: Returns true if the student studies mathematics.
- $G(S_1, S_2)$: Returns true if student S_1 has a higher grade than student S_2 .

Express the following statements using formal mathematical logic.

- (a) [1 point] Every student who studies computer science also studies mathematics.
- (b) [1 point] There exists a student who studies both computer science and mathematics.
- (c) [1 point] A student will study mathematics unless they are studying computer science.
- (d) [2 points] If a student studies both computer science and mathematics, they will have a higher grade than a student who studies only computer science.

Solution.

- (a) $\forall S, C(S) \rightarrow M(S)$
- (b) $\exists S, C(S) \wedge M(S)$
- (c) $\forall S, \sim C(S) \rightarrow M(S)$
- (d) $\forall S_1, S_2, (C(S_1) \wedge M(S_1) \wedge C(S_2) \wedge \sim M(S_2)) \rightarrow G(S_1, S_2)$

3. [5 points] Consider a scenario where an animal is represented by A , and the following predicates are given:

- $C(A)$: Returns true if the animal is a carnivore.
- $H(A)$: Returns true if the animal is herbivorous.
- $F(A)$: Returns true if the animal is a flying species.
- $L(A_1, A_2)$: Returns true if animal A_1 is larger than animal A_2 .

Express the following statements using formal mathematical logic.

- (a) [1 point] All carnivores are not herbivorous.
- (b) [1 point] There exists an animal that is both carnivorous and herbivorous.
- (c) [1 point] Any animal that is not carnivorous is herbivorous or flying.
- (d) [2 points] If an animal is both carnivorous and flying, it is larger than an animal that is herbivorous but not flying.

Solution.

- (a) $\forall A, C(A) \rightarrow \sim H(A)$
- (b) $\exists A, C(A) \wedge H(A)$
- (c) $\forall A, \sim C(A) \rightarrow (H(A) \vee F(A))$
- (d) $\forall A_1, A_2, (C(A_1) \wedge F(A_1) \wedge H(A_2) \wedge \sim F(A_2)) \rightarrow L(A_1, A_2)$

4. [5 points] Determine whether the following statements are true. Assume that $\mathbb{Q}$ and $\mathbb{I}$ are the sets of rational and irrational numbers, respectively.

- (a) [1 point] $\forall x, y \in \mathbb{Z}, xy \in \mathbb{Z}$
- (b) [1 point] $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x > y$
- (c) [1 point] $\forall x, y \in \mathbb{Z}, x/y \in \mathbb{Q}$
- (d) [1 point] $\forall x, y \in \mathbb{I}, x + y \in \mathbb{I}$
- (e) [1 point] $\forall x, y \in \mathbb{Z}, |x + y| \leq |x| + |y|$

Solution.

- (a) True. Integers are closed under multiplication.
- (b) True. There always exists a smaller integer.
- (c) False. Suppose $y = 0$.
- (d) False. $\sqrt{2} + (-\sqrt{2}) = 0$.
- (e) True. This is the triangle inequality.

[Proof Techniques.]

1. [0 points] Which of these are actual mathematical proof techniques?

- Proof by assuming the problem statement
- Proof by intimidation
- Proof by shouting
- Proof by not listening
- Proof by repeatedly saying the given reasoning is a proof
- Proof by showing example(s) for a universal statement
- Proof by trivializing the problem
- Proof by giving the correct answer without reasoning

Solution. None of these.

2. [0 points] Write a beautiful poem on proof techniques.

Solution. ChatGPT: Here it is.

The Proof's Pursuit: A Mathematical Tale

In lands of logic, sharp and keen,
Where numbers dance and minds are seen,
A theorem stood, so bold, so grand—
Yet proofless still, like castles of sand.

With chalk in hand, the wizards came,
Each hoping they could stake their claim.
A contradiction! A bijection!
A proof by cases—pure perfection!

First came Induction, noble and proud,
"I'll prove for one, then for the crowd!
If n is true, and $n + 1$,
The proof is done! Oh, what fun!"

Then came Contradiction's art,
A devious play, a cunning start.
"Assume I'm wrong, let chaos reign,
Oh dear! The logic breaks in twain!"

Contraposition took its turn,
With logic crisp and truth to burn.
"If not B, then not A too,
The proof holds firm, the claim is true!"

Next, Direct Proof took the stage,
With elegance upon the page.

"A to B, so crisp and clean,
No tricks, no traps—just what is seen."

Proof by Cases made its stand,
With many paths already planned.
"If A is this, or A is that,
We check them all—now that's a fact!"

Reduction came, so sly and slick,
"Transform it here—just one neat trick!
A hard old proof made easy now,
Who needs brute force? Just take a bow!"

Then One-to-One, so sharp and bright,
Mapped all things left to right.
"A bijection, neat and true,
Two sets the same—just look, you'll see too!"

The audience gasped, the scribes took note,
The theorem stood, as if it wrote,
"I stand, I stand! I am complete!"
The mathematicians rose to their feet.

Yet lurking still in shadow's guise,
A figure grinned with knowing eyes.
From Gödel's lips, a whisper grew:
"Not all theorems can be proved!"

And so they sighed, both sad and wise,
For proofs may shine, but truth still hides.
Yet undeterred, they took their quills,
For math rewards the steadfast wills.

So let us strive, with joy and glee,
For proofs are puzzles, wild and free!
Through logic's maze, we twist and wind—
Who knows what wonders we may find?

3. [5 points] Consider the following propositions.

- (i) Prove that if n is odd, then n^{10} is odd.
- (ii) Prove that if $3|n$, then $6|n$.
- (iii) Prove that $\sqrt{2}$ is irrational.
- (iv) Prove that if n is an integer, then n^2 is even.
- (v) Prove that if $a|b$ and $b|c$, then $a|c$.

Mention which type(s) of proof could be easily applied as seen in the lectures to solve each of the propositions listed above. There may be multiple correct answers

for each proposition.

- (a) Direct Proof
- (b) Proof by Contradiction
- (c) Proof by Contrapositive
- (d) Proof by Division into Cases
- (e) Disproof by Counterexample

Solution.

- (i) (a), (b)
- (ii) (e)
- (iii) (b)
- (iv) (d)
- (v) (a), (b)

4. [5 points] Prove that the sum of any two integers with the same parity (either both even or both odd) results in an even number.

Solution.

Case 1: Both integers are even. Let a and b be two even integers. Then

$$\begin{aligned} a + b &= 2m + 2n \quad (\text{defn. of even, } a = 2m, b = 2n) \\ &= 2(m + n) \quad (\text{factor out 2}) \\ &= 2p \quad (p = m + n \text{ and addition is closed on integers}) \\ &= \text{even} \quad (\text{defn. of even}) \end{aligned}$$

Case 2: Both integers are odd. Let a and b be two odd integers. Then

$$\begin{aligned} a + b &= (2m + 1) + (2n + 1) \quad (\text{defn. of odd, } a = 2m + 1, b = 2n + 1) \\ &= 2m + 2n + 2 \quad (\text{simplify the terms}) \\ &= 2(m + n + 1) \quad (\text{factor out 2}) \\ &= 2p \quad (p = m + n + 1 \text{ and addition is closed on integers}) \\ &= \text{even} \quad (\text{defn. of even}) \end{aligned}$$

5. [5 points] Prove that if n^{100} is odd, then n is odd.

Solution. We will prove the statement by contrapositive.

Contrapositive: If n is even, then n^{100} is even.

Proof: Assume n is even. Then $n = 2k$ for some integer k . Now, consider n^{100} :

$$\begin{aligned} n^{100} &= (2k)^{100} \quad (\text{since } n = 2k) \\ &= 2^{100}k^{100} \quad (\text{simplify the expression}) \end{aligned}$$

Since 2^{100} is a power of 2, it is clearly divisible by 2, so n^{100} is even.

Thus, by contrapositive, if n^{100} is odd, then n must be odd. This completes the proof.

6. [5 points] Prove that the sum of any two prime numbers greater than 2 is always even.

Solution.

Let p_1 and p_2 be prime numbers greater than 2. Then

$$p_1 = 2k_1 + 1 \text{ and } p_2 = 2k_2 + 1 \quad (\text{defn. of odd, since primes greater than 2 are odd})$$

$$\begin{aligned}
p_1 + p_2 &= (2k_1 + 1) + (2k_2 + 1) \\
&= 2k_1 + 2k_2 + 2 \quad (\text{simplifying the terms}) \\
&= 2(k_1 + k_2 + 1) \quad (\text{factor out 2}) \\
&= 2k_3 \quad (\text{where } k_3 = k_1 + k_2 + 1) \\
&= \text{even} \quad (\text{defn. of even})
\end{aligned}$$

7. [5 points] Prove the following statement using two different methods: (1) by contrapositive and (2) by contradiction.

"If the product of two integers is odd, then both integers must be odd."

Solution.

Proof 1: Proof by Contrapositive

Contrapositive: If at least one of the two integers is even, their product is even. Assume at least one integer is even. Let $a = 2k$ for some integer k .

$$\begin{aligned}
a \times b &= (2k) \times b \quad (\text{defn. of even, } a = 2k) \\
&= 2(k \times b) \quad (\text{factor out 2}) \\
&= 2m \quad (\text{where } m = k \times b) \\
&= \text{even} \quad (\text{defn. of even})
\end{aligned}$$

Thus, if at least one integer is even, their product is even.

By contrapositive, if $a \times b$ is odd, both integers are odd.

Proof 2: Proof by Contradiction

Negation: The product of two integers is odd and one of the two integers is even. Let a and b be two integers.

Assume $a \times b$ is odd, but one integer is even.

Let $a = 2k$. Then

$$\begin{aligned}
a \times b &= (2k) \times b \quad (\text{defn. of even, } a = 2k) \\
&= 2(k \times b) \quad (\text{factor out 2}) \\
&= 2m \quad (\text{where } m = k \times b) \\
&= \text{even} \quad (\text{defn. of even})
\end{aligned}$$

Contradiction. Thus, both integers must be odd.

8. [5 points] Prove that if a and b are integers such that $a \mid b$ and $b \mid c$, then $a \mid c$.

Solution.

Given $a \mid b$ and $b \mid c$, there exist integers k_1 and k_2 such that:

$$b = a \times k_1 \quad (\text{defn. of } a \mid b)$$

$$c = b \times k_2 \quad (\text{defn. of } b \mid c)$$

Substitute $b = a \times k_1$ into $c = b \times k_2$ to get:

$$\begin{aligned}
c &= (a \times k_1) \times k_2 \\
&= a \times (k_1 \times k_2) \quad (\text{associativity of multiplication}) \\
&= a \times k_3 \quad (\text{Let } k_3 = k_1 \times k_2 \text{ and } k_3 \text{ is an integer})
\end{aligned}$$

Therefore, $a \mid c$.

9. [5 points] Consider the following statements. For each statement, either prove it or provide a counterexample.

- (a) [2 points] The product of two irrational numbers is always irrational.
- (b) [3 points] The sum of two positive irrational numbers is always irrational.

Solution.

(a) False. Consider $\sqrt{2} \cdot \sqrt{2} = 2$.

(b) False. Consider $\sqrt{2} + (2 - \sqrt{2}) = 2$

[Sequences.]

1. [5 points] Prove by induction that the following statement is true for all integers $n \geq 1$:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

Solution. Let $P(n) : \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.

Base case: For $n = 1$,

$$P(1) : \sum_{i=1}^1 i^2 = 1 = \frac{1 \times 2 \times 3}{6}$$

Thus, the base case holds.

Inductive step: Assume $P(k)$ is true for some $k \geq 1$. We need to prove $P(k+1)$:

LHS of $P(k+1)$:

$$= \sum_{i=1}^{k+1} i^2 = \sum_{i=1}^k i^2 + (k+1)^2 \quad (\text{by definition of summation})$$

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad (\text{since } P(k) \text{ is assumed to be true})$$

$$= \frac{(k+1)(2k^2+7k+6)}{6} \quad (\text{simplify and combine terms})$$

$$= \frac{(k+1)(k+2)(2k+3)}{6} \quad (\text{factor the expression})$$

$$= \text{RHS of } P(k+1) \text{ Thus, the inductive step holds.}$$

Therefore, by mathematical induction, $P(n)$ is true for all $n \geq 1$.

2. [5 points] Prove by induction that

$$\prod_{i=1}^n 2i = 2^n \cdot n!$$

for all $n \in \mathbb{N}$.

Solution. Let $P(n) : \prod_{i=1}^n 2i = 2^n \cdot n!$.

Base case: For $n = 1$, the left-hand side is:

$$2 = 2^1 \cdot 1! = 2$$

Thus, $P(1)$ holds.

Inductive step: Assume $P(k)$ is true for some $k \geq 1$. We must prove $P(k+1)$:

LHS of $P(k+1)$:

$$= (\prod_{i=1}^k 2i) \cdot 2(k+1) \quad (\text{by definition of product})$$

$$= (2^k \cdot k!) \cdot 2(k+1) \quad (\text{by inductive hypothesis})$$

$$= 2^{k+1} \cdot (k+1)! \quad (\text{simplify})$$

$$= \text{RHS of } P(k+1).$$

Thus, the inductive step holds.

Therefore, by mathematical induction, $P(n)$ holds for all $n \in \mathbb{N}$.

3. [5 points] Prove by induction that if

$$a_n = \frac{a_{n-1} \cdot a_{n-2}}{a_{n-1} + a_{n-2}},$$

with initial conditions $a_1 = 2$ and $a_2 = 1$, then for all $n \in \mathbb{N}$,

$$a_n \leq \frac{2}{n}.$$

Solution. Let $P(n) : a_n \leq \frac{2}{n}$. We will prove $P(n)$ by induction for all $n \in \mathbb{N}$.

Base case:

For $n = 1$, $a_1 = 2 \leq \frac{2}{1}$. (True)

For $n = 2$, $a_2 = 1 \leq \frac{2}{2}$. (True)

Thus, $P(1)$ and $P(2)$ hold.

Inductive step: Assume $P(i)$ holds true for all $i \in [1, k]$ for some $k \geq 2$. We need to show that $P(k+1)$ holds.

LHS of $P(k+1)$:

$$\begin{aligned} &= a_{k+1} \\ &= \frac{a_k \cdot a_{k-1}}{a_k + a_{k-1}} \quad (\text{from the recurrence relation}) \\ &\leq \frac{\frac{2}{k} \cdot \frac{2}{k-1}}{\frac{2}{k} + \frac{2}{k-1}} \quad (P(k) : a_k \leq \frac{2}{k} \text{ and } P(k-1) : a_{k-1} \leq \frac{2}{k-1} \text{ are true}) \\ &= \frac{\frac{4}{k(k-1)}}{\frac{2k+2(k-1)}{k(k-1)}} = \frac{4}{4k-2} = \frac{2}{2k-1} \leq \frac{2}{k+1} \quad (\text{simplify}) \\ &= \text{RHS of } P(k+1) \end{aligned}$$

Therefore, $a_{k+1} \leq \frac{2}{k+1}$, and $P(k+1)$ holds.

By induction, $P(n)$ is true for all $n \in \mathbb{N}$. Therefore, $a_n \leq \frac{2}{n}$ for all $n \in \mathbb{N}$.

4. [5 points] Prove by induction that if

$$a_n = \sqrt{a_{n-3} + a_{n-2} + a_{n-1}} \quad \text{for all } n \geq 4$$

and $a_1 = 0$, $a_2 = 1$, $a_3 = 2$, then for all $n \in \mathbb{N}$, $a_n < 3$.

Solution. Let $P(n) : a_n < 3$. We will prove $P(n)$ by induction for all $n \in \mathbb{N}$.

Base cases

For $n = 1$, $a_1 = 0 < 3$.

For $n = 2$, $a_2 = 1 < 3$.

For $n = 3$, $a_3 = 2 < 3$.

Thus, $P(1)$, $P(2)$, and $P(3)$ hold.

Inductive step: Assume $P(i)$ holds true for all $i \in [1, k]$ for some $k \geq 3$. We need to show that $P(k+1)$ holds.

LHS of $P(k+1)$

$$\begin{aligned} &= a_{k+1} \\ &= \sqrt{a_{k-2} + a_{k-1} + a_k} \quad (\text{from the recurrence relation}) \\ &< \sqrt{3 + 3 + 3} \quad (P(k) : a_k < 3, P(k-1) : a_{k-1} < 3, P(k-2) : a_{k-2} < 3 \text{ are true}) \\ &= \sqrt{9} = 3 \\ &= \text{RHS of } P(k+1) \end{aligned}$$

Therefore, $a_{k+1} < 3$, and $P(k+1)$ holds.

By induction, $P(n)$ is true for all $n \in \mathbb{N}$. Therefore, $a_n < 3$ for all $n \in \mathbb{N}$.

[Sets.]

1. [5 points] Given that $A = \{1, 3, 5, 8\}$, $B = \{2, 3, 4, 5\}$, $C = \{1, 4, 8\}$ and $D = \{1\}$.
- (a) [1 point] What is the value of $|A - B|$?
 - (b) [1 point] What is the value of $|(A \cup B) \cap (C \cup D)|$?
 - (c) [1 point] What is $(A - B) \cup (C - D)$?
 - (d) [2 points] What is $\mathcal{P}(A - B) \cap \mathcal{P}(C - D)$?

Solution.

- (a) 2
 - (b) 3
 - (c) $\{1, 4, 8\}$
 - (d) $\{\emptyset, \{8\}\}$
2. [5 points] Prove that if $A \subseteq B$, then $A - B = \emptyset$.

Solution.

We prove by contradiction.

Negation: Suppose that $A \subseteq B$ and $A - B$ is not an empty set

Suppose $x \in A - B$.

$\implies x \in A$ and $x \notin B$ (by definition of set difference)

$\implies x \in B$ (since $A \subseteq B$)

Contradiction since $x \in B$ and $x \notin B$.

$\implies A - B = \emptyset$.

3. [5 points] Prove that if $A \cup B = U$ and $A \cap B = \emptyset$, then $(A \cup C) \cap (B \cup C) = C$.

Solution.

We will prove $(A \cup C) \cap (B \cup C) = C$ in two parts.

Part 1: $(A \cup C) \cap (B \cup C) \subseteq C$.

Let $x \in (A \cup C) \cap (B \cup C)$.

$\implies x \in A \cup C$ and $x \in B \cup C$ (by definition of intersection).

$\implies (x \in A \text{ or } x \in C) \text{ and } (x \in B \text{ or } x \in C)$ (by definition of union)

$\implies ((x \in A \text{ and } x \in B) \text{ or } x \in C)$ (by distributive law)

$\implies x \in C$ (since $A \cap B = \emptyset$, so $x \notin A$ and $x \notin B$).

Thus, $x \in C$.

Therefore, $(A \cup C) \cap (B \cup C) \subseteq C$.

Part 2: $C \subseteq (A \cup C) \cap (B \cup C)$.

Let $x \in C$.

$\implies x \in A \cup C$ and $x \in B \cup C$ (by definition of union).

$\implies x \in (A \cup C) \cap (B \cup C)$.

$\implies C \subseteq (A \cup C) \cap (B \cup C)$.

Since both inclusions hold, we conclude that $(A \cup C) \cap (B \cup C) = C$.

[Functions.]

1. [5 points] Let $A = \{0, 2, 4, 6, 8\}$ and $B = \{1, 2, 3, 4\}$. Answer the following:
 - (a) [1 point] Calculate $|A \cup B| - |A \cap B|$.
 - (b) [1 point] Does there exist a one-to-one correspondence function $f : A \rightarrow B$?
If so, what is a possible f ?
 - (c) [1 point] Find the number of one-to-one functions $g : A \rightarrow B$.
 - (d) [2 points] Let $f : A \rightarrow B$ and $g : B \rightarrow A$. Is it possible that $f \circ g = g \circ f$?

Solution.

- (a) 5.
 - (b) There does not exist such an f .
 - (c) 0 functions.
 - (d) This is not possible because f, g are not one-to-one correspondences so they cannot be inverses of one another.
2. [5 points] Let A and B be two finite sets with sizes $|A| = n$ and $|B| = m$. Answer the following questions about functions between A and B :
 - (a) [2 points] Prove that a one-to-one function $f : A \rightarrow B$ exists if $n \leq m$.
 - (b) [2 points] Prove that an onto function $f : A \rightarrow B$ exists if $n \geq m$.
 - (c) [1 point] Show that a one-to-one correspondence $f : A \rightarrow B$ exists if $n = m$.

Solution.

- (a) If $n > m$, the pigeonhole principle implies that at least two elements of A would map to the same element of B , violating the one-to-one condition. Therefore, $n \leq m$ is necessary.
 - (b) One can visually show that if $n < m$, there would be elements of B with no preimage, violating the onto condition. Therefore, $n \geq m$ is necessary.
 - (c) Combining the results from parts (a) and (b), this requires $n = m$. For f to be one-to-one, $n \leq m$. For f to be onto, $n \geq m$. Together, $m \leq n \leq m \implies n = m$.
3. [5 points] Let $f : \mathbb{Z}^+ \rightarrow \mathbb{O}^+$, where $\mathbb{O}^+$ represents the set of positive odd numbers, be defined as $f(x) = 2x - 1$. Prove that f is a one-to-one correspondence.

Solution.

To prove that f is a one-to-one correspondence, we need to show that f is both one-to-one and onto.

1. One-to-One: A function f is one-to-one $\forall a, b \in \mathbb{Z}^+$ if $f(a) = f(b)$ then $a = b$.

Suppose $f(a) = f(b)$. Then, $2a - 1 = 2b - 1$.

Simplifying:

$$2a = 2b \implies a = b \implies f \text{ is one-to-one.}$$

2. Onto: A function f is onto if, for every $y \in \mathbb{O}^+$, there exists an $x \in \mathbb{Z}^+$ such that $f(x) = y$. Let $y \in \mathbb{O}^+$. Since $\mathbb{O}^+ = \{1, 3, 5, 7, \dots\}$, we can write $y = 2x - 1$. Solving for x :

$$x = \frac{y+1}{2}.$$

Since y is odd, $y+1$ is even, and $\frac{y+1}{2} \in \mathbb{Z}^+$. Hence, $f(x) = y$, proving f is onto.

Conclusion: Since f is both one-to-one and onto, f is a one-to-one correspondence.

4. [5 points] Sort the following sets by non-decreasing cardinality using $<$ and $=$ symbols:

$$\emptyset, \mathbb{N}, \mathcal{P}(\mathbb{N}), \mathcal{P}(\mathcal{P}(\mathbb{N})), \mathbb{Q}, \mathbb{R}, \mathcal{P}(\mathbb{R}), \{1\}$$

Solution.

$$|\emptyset| < |\{1\}| < |\mathbb{N}| = |\mathbb{Q}| < |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| = |\mathcal{P}(\mathbb{R})|$$

5. [5 points] Consider $f : \mathbb{R}^* \rightarrow \mathbb{R}^*$, where $\mathbb{R}^* = \mathbb{R}^+ \cup \{0\}$ is the set of all nonnegative real numbers, such that f is a mapping from pounds to kilograms. A kilogram of mass in the International System of Units (SI) represents 2.2046226218 pounds. Is f a one-to-one correspondence? Is f a one-to-one correspondence? If yes, give a formal proof.

Solution. Given that 1 kilogram = 2.2046226218 pounds, we can express the function f as:

$$f(x) = \frac{x}{2.2046226218}$$

where x is the mass in pounds. The function maps from the set of nonnegative real numbers (including zero) to the set of nonnegative real numbers. Since $2.2046226218 > 0$, for any $x \geq 0$, $f(x)$ will also be nonnegative. Therefore, $f(x) \in \mathbb{R}^*$ for all $x \in \mathbb{R}^*$. Thus, f is well-defined.

1. One-to-One:

To show that f is injective, we need to prove that if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

Assume $f(x_1) = f(x_2)$:

$$\frac{x_1}{2.2046226218} = \frac{x_2}{2.2046226218}$$

Multiplying both sides by 2.2046226218 (which is positive and does not change the equality) gives:

$$x_1 = x_2$$

Thus, f is injective.

2. Onto:

To show that f is surjective, we need to prove that for every $y \in \mathbb{R}^*$, there exists an $x \in \mathbb{R}^*$ such that $f(x) = y$.

Let y be any nonnegative real number. We want to find x such that:

$$f(x) = y \implies \frac{x}{2.2046226218} = y$$

Solving for x gives:

$$x = 2.2046226218y$$

Since $y \geq 0$, it follows that $x = 2.2046226218y \geq 0$. Therefore, for every $y \in \mathbb{R}^*$, we can find an $x \in \mathbb{R}^*$ such that $f(x) = y$. Thus, f is surjective.

Conclusion:

Since f is both injective and surjective, we conclude that f is a one-to-one correspondence (bijection) from $\mathbb{R}^*$ to $\mathbb{R}^*$. Thus, the function f is a one-to-one correspondence.

6. [5 points] Consider the set of numbers of the form k^2 or 2^k , where $k \in \mathbb{N}$. Prove that the set is countable by coming up with a one-to-one correspondence function and showing the arrow diagram.

Solution.

To prove that the set consisting of numbers of the form k^2 or 2^k , where $k \in \mathbb{N}$, is countable, we will first define the set clearly and then establish a one-to-one correspondence function.

Step 1: Define the Set

Let S be the set defined as follows:

$$S = \{k^2 \mid k \in \mathbb{N}\} \cup \{2^k \mid k \in \mathbb{N}\}$$

This means that S contains all perfect squares and all powers of 2, where k is a natural number.

Step 2: List the Set Elements

Perfect Squares: The elements of the form k^2 for $k = 1, 2, 3, \dots$ are:

$$1^2 = 1, 2^2 = 4, 3^2 = 9, 4^2 = 16, 5^2 = 25, 6^2 = 36, 7^2 = 49, 8^2 = 64, 9^2 = 81, \dots$$

Powers of 2: The elements of the form 2^k for $k = 1, 2, 3, \dots$ are:

$$2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32, 2^6 = 64, 2^7 = 128, 2^8 = 256, \dots$$

Common elements:

4 appears as 2^2 and 2^2 .

16 appears as 4^2 and 2^4 .

64 appears as 8^2 and 2^6 .

so on...

So, essentially, we will need to eliminate all numbers of the form 2^{even} to avoid double-counting.

Step 3: Create a One-to-One Correspondence

To show that the set S is countable, we can create a function that maps each natural number k to a unique element in S .

Define a function $f : \mathbb{N} \rightarrow S$ as follows:

$$f(n) = \begin{cases} 2^n & \text{if } n \text{ is odd,} \\ (n/2)^2 & \text{if } n \text{ is even.} \end{cases}$$

Step 4: Diagram of the Function

n	$f(n)$
1	2^1
2	1^2
3	2^3
4	2^2
5	2^5
6	3^2
7	2^7
8	4^2
$\vdots$	$\vdots$

[Relations]

1. [10 points] Consider finite sets A and B such that $|A| = m, |B| = n$.
- (a) [1 point] How are $|A \cup B|$ and $|A \cap B|$ related to each other?
 - (b) [1 point] What is $|\mathcal{P}(A \times B)|$?
 - (c) [2 points] How many relations $R : A \rightarrow B$ exist?
 - (d) [2 points] How many functions $f : A \rightarrow B$ exist?
 - (e) [2 points] How many one-to-one functions $f : A \rightarrow B$ exist?
 - (f) [2 points] How many one-to-one correspondences functions $f : A \rightarrow B$ exist?

Solution.

- (a) $|A \cup B| = |A| + |B| - |A \cap B| = m + n - |A \cap B|$
- (b) 2^{mn}
($|A \times B| = mn$. The number of subsets of mn elements is 2^{mn} .)
- (c) 2^{mn}
(Total number of binary relations from A to B is the total number of subsets of $A \times B$)
- (d) n^m
(First input element can be mapped to any of n output elements, so on till the m th input element)
- (e)
$$\begin{cases} 0 & \text{if } m > n, \\ n \times (n-1) \times (n-2) \times \cdots \times (n-m+1) & \text{if } m \leq n. \end{cases}$$

(First input element can be mapped to any of n output elements, second input element can be mapped to any of remaining $(n-1)$ output elements, so on till the m th input element can be mapped to any of $(n-m+1)$ output elements)
- (f)

$$\begin{cases} 0 & \text{if } m \neq n, \\ m! & \text{if } m = n. \end{cases}$$

(First input element can be mapped to any of m output elements, second input element can be mapped to any of remaining $(m-1)$ output elements, so on till the m th input element can be mapped to the only remaining 1 output element)

2. [5 points] Let R be a relation on the set of integers $\mathbb{Z}$ defined by $a R b$ if and only if $a - b$ is divisible by 3. Check if R is an equivalence relation and describe the equivalence classes of R if it is.

Solution.

1. Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$ is divisible by 3. Hence, $a R a$ holds.

2. Symmetric: If $a R b$, then $a - b = 3k$ for some $k \in \mathbb{Z}$. Thus, $b - a = -3k$, which is divisible by 3. Hence, $b R a$ holds.

3. Transitive: If $a R b$ and $b R c$, then $a - b = 3k$ and $b - c = 3m$ for some $k, m \in \mathbb{Z}$. Adding these gives $a - c = 3(k + m)$, which is divisible by 3. Hence, $a R c$.

4. Equivalence Classes: The equivalence classes are the sets of integers that leave a remainder of 0, 1, and 2 when divided by 3, as shown below:

$$[0] = \{\dots, -3, 0, 3, 6, \dots\}, \quad [1] = \{\dots, -2, 1, 4, 7, \dots\}, \quad [2] = \{\dots, -1, 2, 5, 8, \dots\}.$$

3. [5 points] Let R be a relation on the set of all strings over the alphabet $\{a, b\}$ defined by $x R y$ if and only if x and y have the same length. Check if R is an equivalence relation and describe the equivalence classes of R if it is.

Solution.

1. Reflexive: For any string x , we have $|x| = |x|$, so $x R x$ holds.

2. Symmetric: If $x R y$, then $|x| = |y|$. Thus, $|y| = |x|$, so $y R x$ holds.

3. Transitive: If $x R y$ and $y R z$, then $|x| = |y|$ and $|y| = |z|$. Thus, $|x| = |z|$, so $x R z$ holds.

4. Equivalence Classes: The equivalence classes are the sets of strings with the same length. For any non-negative integer n , the equivalence class of strings of length n is:

$$[n] = \{x \mid x \text{ is a string over } \{a, b\} \text{ with length } n\}$$

4. [5 points] Let R be a relation on $\mathbb{R}$ defined by $a R b$ if and only if $a^2 = b^2$. Check if R is an equivalence relation and describe the equivalence classes of R if it is.

Solution.

1. Reflexive: For any $a \in \mathbb{R}$, we have $a^2 = a^2$, so $a R a$ holds.

2. Symmetric: If $a R b$, then $a^2 = b^2$. Thus, $b^2 = a^2$, so $b R a$ holds.

3. Transitive: If $a R b$ and $b R c$, then $a^2 = b^2$ and $b^2 = c^2$. Thus, $a^2 = c^2$, so $a R c$ holds.

4. Equivalence Classes: The equivalence classes are the sets of real numbers with the same absolute value. For any non-negative real number r , the equivalence class of r is:

$$[r] = \{x \in \mathbb{R} \mid |x| = r\} = \{-r, r\} \quad (\text{if } r \neq 0), \text{ and } [0] = \{0\}.$$

5. [5 points] Let R be a relation on the set of all people, where $a R b$ if and only if a and b have the same birth year. Check if R is an equivalence relation and describe the equivalence classes of R if it is.

Solution.

1. Reflexive: For any person a , the relation holds because a has the same birth year as a . Thus, $a R a$ holds.

2. Symmetric: If $a R b$, then a and b have the same birth year. Since having the same birth year is a mutual property, $b R a$ also holds.

3. Transitive: If $a R b$ and $b R c$, then a and b have the same year, and b and c have the same year. Thus, a and c must also have the same birth year, so $a R c$ holds.

4. Equivalence Classes: The equivalence classes of R are the sets of people born in the same year. For any year y , the equivalence class of people born in year y is:

$$[y] = \{a \mid \text{person } a \text{ is born in year } y\}.$$

6. [5 points] Let R be a relation on the set of all points in the plane $\mathbb{R}^2$ defined by $(x_1, y_1) R (x_2, y_2)$ if and only if $x_1 = x_2$ or $y_1 = y_2$. Check if R is an equivalence relation and describe the equivalence classes of R if it is.

Solution.

1. Reflexive: For any point (x_1, y_1) , we have $(x_1, y_1) R (x_1, y_1)$ since $x_1 = x_1$ and $y_1 = y_1$. Thus, the relation is reflexive.

2. Symmetric: If $(x_1, y_1) R (x_2, y_2)$, then either $x_1 = x_2$ or $y_1 = y_2$. If $x_1 = x_2$, then clearly $x_2 = x_1$. If $y_1 = y_2$, then clearly $y_2 = y_1$. Thus, the relation is symmetric.

3. Not Transitive: Consider $(1, 1), (1, 2), (2, 2)$. $(1, 1) R (1, 2)$, $(1, 2) R (2, 2)$, but $(1, 1) \bar{R} (2, 2)$. Therefore, not transitive.

This is NOT an equivalence relation.

7. [5 points] Find the units digit of 7^{100} .

Solution.

To find the units digit of 7^{100} , observe the pattern of the units digits of powers of 7:

$$\begin{aligned} 7^1 &= 7 \quad (\text{units digit } 7), \\ 7^2 &= 49 \quad (\text{units digit } 9), \\ 7^3 &= 343 \quad (\text{units digit } 3), \\ 7^4 &= 2401 \quad (\text{units digit } 1). \end{aligned}$$

The units digits repeat in a cycle of 4: 7, 9, 3, 1. To find the units digit of 7^{100} , divide 100 by 4:

$$100 \div 4 = 25 \quad \text{remainder } 0.$$

Since the remainder is 0, the units digit of 7^{100} corresponds to the units digit of 7^4 , which is 1. Thus, the units digit of 7^{100} is 1.

8. [5 points] Find the units digit of 3^{50} .

Solution.

To find the units digit of 3^{50} , observe the pattern of the units digits of powers of 3:

$$3^1 = 3 \quad (\text{units digit } 3),$$

$$3^2 = 9 \quad (\text{units digit } 9),$$

$$3^3 = 27 \quad (\text{units digit } 7),$$

$$3^4 = 81 \quad (\text{units digit } 1).$$

The units digits repeat in a cycle of 4: 3, 9, 7, 1. To find the units digit of 3^{50} , divide 50 by 4:

$$50 \div 4 = 12 \quad \text{remainder } 2.$$

Since the remainder is 2, the units digit of 3^{50} corresponds to the units digit of 3^2 , which is 9. Thus, the units digit of 3^{50} is $\boxed{9}$.

9. [5 points] Find the units digit of 12^{1234} .

Solution.

To find the units digit of 12^{1234} , observe the pattern of the units digits of powers of 12. The units digits of powers of 12 follow the same pattern as powers of 2:

$$12^1 = 12 \quad (\text{units digit } 2),$$

$$12^2 = 144 \quad (\text{units digit } 4),$$

$$12^3 = 1728 \quad (\text{units digit } 8),$$

$$12^4 = 20736 \quad (\text{units digit } 6).$$

The units digits repeat in a cycle of 4: 2, 4, 8, 6. To find the units digit of 12^{1234} , divide 1234 by 4:

$$1234 \div 4 = 308 \quad \text{remainder } 2.$$

Since the remainder is 2, the units digit of 12^{1234} corresponds to the units digit of 12^2 , which is 4. Thus, the units digit of 12^{1234} is $\boxed{4}$.

10. [5 points] Find the units digit of 2^{987} .

Solution.

To find the units digit of 2^{987} , observe the pattern of the units digits of powers of 2:

$$2^1 = 2 \quad (\text{units digit } 2),$$

$$2^2 = 4 \quad (\text{units digit } 4),$$

$$2^3 = 8 \quad (\text{units digit } 8),$$

$$2^4 = 16 \quad (\text{units digit } 6).$$

The units digits repeat in a cycle of 4: 2, 4, 8, 6. To find the units digit of 2^{987} , divide 987 by 4:

$$987 \div 4 = 246 \quad \text{remainder } 3.$$

Since the remainder is 3, the units digit of 2^{987} corresponds to the units digit of 2^3 , which is 8. Thus, the units digit of 2^{987} is $\boxed{8}$.

11. [5 points] Find the units digit of 9^{999} .

Solution.

To find the units digit of 9^{999} , observe the pattern of the units digits of powers of 9:

$$\begin{aligned}9^1 &= 9 \quad (\text{units digit } 9), \\9^2 &= 81 \quad (\text{units digit } 1).\end{aligned}$$

The units digits repeat in a cycle of 2: 9, 1. To find the units digit of 9^{999} , divide 999 by 2:

$$999 \div 2 = 499 \text{ remainder } 1.$$

Since the remainder is 1, the units digit of 9^{999} corresponds to the units digit of 9^1 , which is 9. Thus, the units digit of 9^{999} is 9.